Relay Logic Circuits

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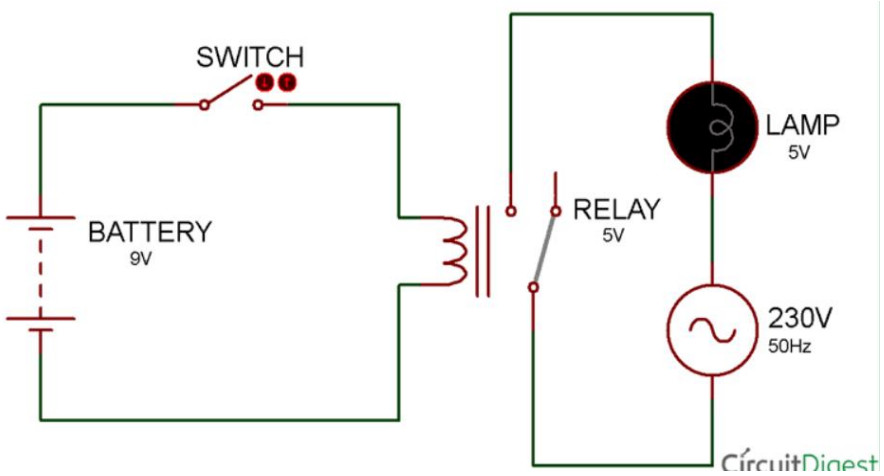
Outline

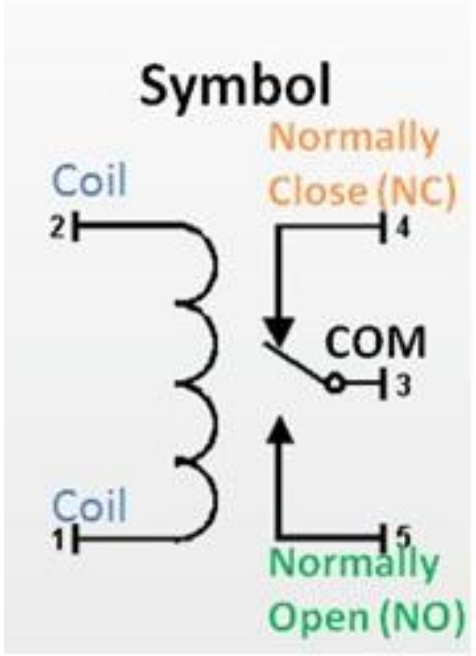
- Relay Basics - Recap
- Control Relays – Latch Circuit
- Relay as a Logic Element
- Logic Circuits with Relays
- Ladder Logic Representation

Relay Basics

A relay is an *electromechanical switch*.

- **Normally Open (NO):** stays open until the coil is energized, then it closes.
- **Normally Closed (NC):** stays closed until the coil is energized, then it open



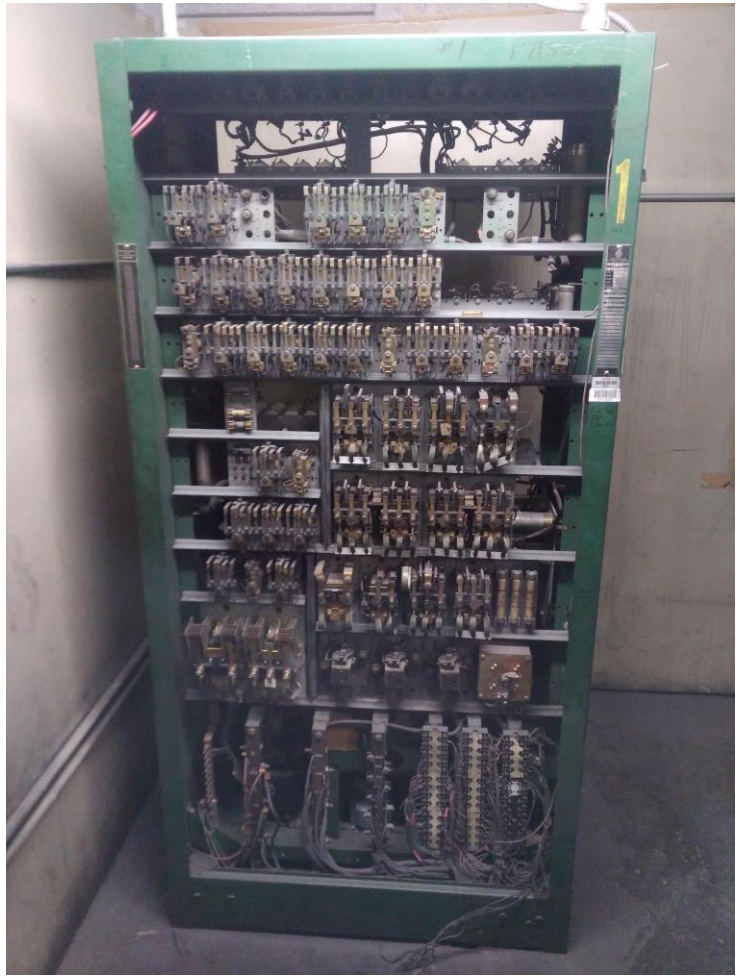


Relay Logic

- Relay Logic is a method of implementing control systems using electromechanical relays to perform logical operation.
- It forms the foundation for early automated control in industrial settings
- Relays can perform basic logic functions by how contacts are wired.

Core Components:

- At its heart, relay logic relies on relays—switches operated by an electromagnet.
- A relay has contacts (normally open or normally closed) that change state when a coil is energized. These can be wired to create circuits that mimic Boolean logic gates/controls.



Relay Logic controller for freight elevator

History

- **Early 1900s**: Relays were used in telephone exchanges to connect calls.
- **1930s–1950s**: Widely used in **industrial automation** to control motors, machines, and safety circuits.
- **Relay Sequencers**: Entire factory processes were automated with relay logic, using latches, timers, and safety interlocks.
- **1960s Onward**: Relays began to be replaced by **Programmable Logic Controllers (PLCs)**, which used the same ladder diagram style so technicians could easily transition

Relay logic = the foundation of **modern automation**.

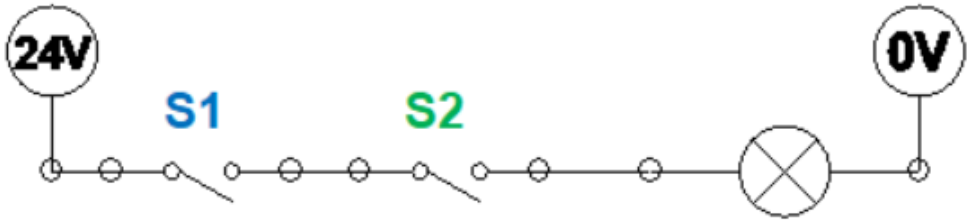
Logic Circuits

- AND: True only if all inputs are true
- OR: True if at least one input is true
- NOT: Inverts the input (0 becomes 1, 1 becomes 0)

AND Logic Circuit

AND Function
all switches must close for current to flow.

Example:
Machine runs only if *Start* **and** *Safety* switches are closed.



Truth table for a 2-input AND gate.

Inputs		Output
A	*B*	*X*
0	0	0
0	1	0
1	0	0
1	1	1

1 = HIGH, 0 = LOW

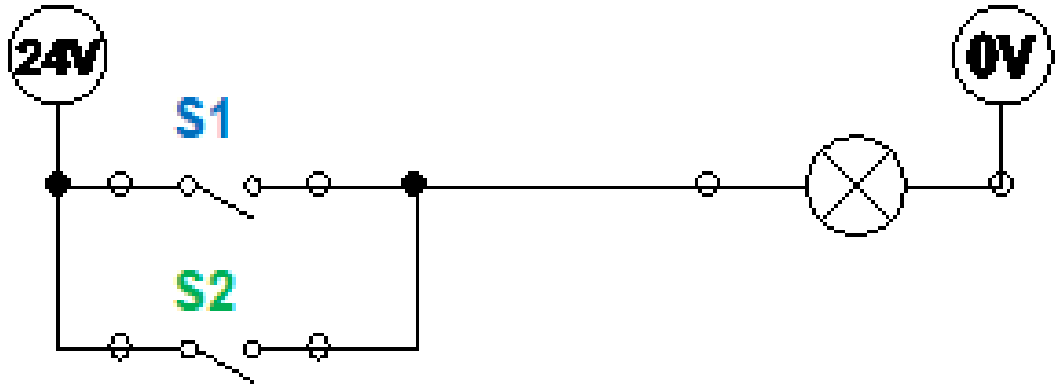
Logic: Output turns ON only if both switches are closed

AND = (S1.S2)

OR Logic Circuit

OR Circuit (Two switches in parallel)
Logic: Output turns ON if either switch is closed.

Example:
Light turns on if *Switch A* **or** *Switch B* is pressed.



A	B	$A + B = X$
0	0	$0 + 0 = 0$
0	1	$0 + 1 = 1$
1	0	$1 + 0 = 1$
1	1	$1 + 1 = 1$

OR = (S1+ S2)

NOT Logic Circuit

Logic: Output is ON when switch is OFF.



Example:
Alarm sounds when *Door Switch* is NOT pressed (door open)

NOT Circuit (Normally Closed contact)

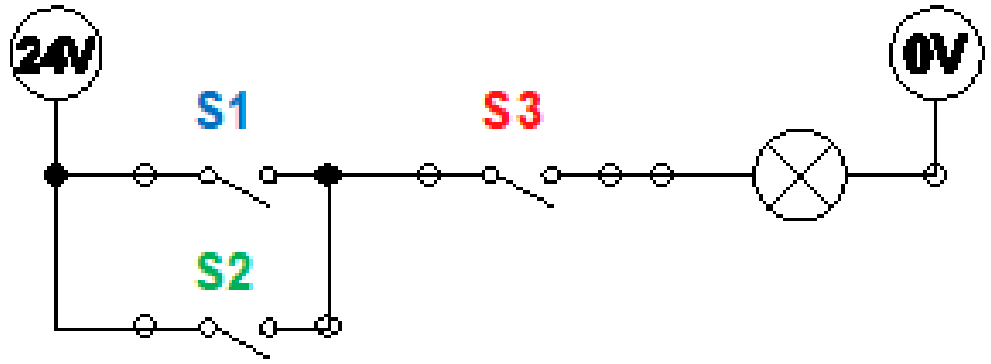
Inverter truth table.

Input	Output
LOW (0)	HIGH (1)
HIGH (1)	LOW (0)

Combination Logic

- Series + parallel contacts allow complex conditions.
- Basis of early relay-based computers and industrial controls.

OR / AND = (S1+ S2) . S3



Relay Logic Circuits

- Graphical representation of relay logic

Other names: Line diagrams/ Ladder logic diagrams

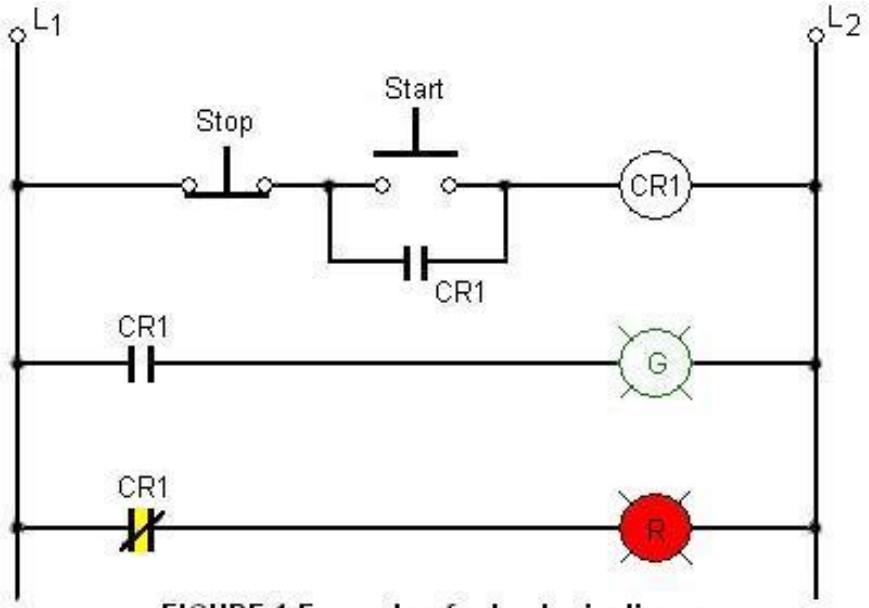


FIGURE 1 Example of relay logic diagram.

A ladder diagram Representation:

- **Two vertical** rails represent the power supply.
- Each **horizontal line**, or rung, represents a control circuit.
- **Power flows** from left to right
- Left side nearest to L1→ inputs (Switches, Sensors)
- Right-side nearest to L2→ Output (Motors, indicators etc.)

Benefits: Easy to read, standard in industrial automation

Relay Logic Circuit Example

Operation:

- The voltage of the control circuit is applied across L1 and L2 (for example, 24 VDC or 120 VAC)
- **CR1 = Control Relay 1**.
- For example, a CR1 NO (normally open) contact will close, and a CR1 NC (normally closed) contact will open.

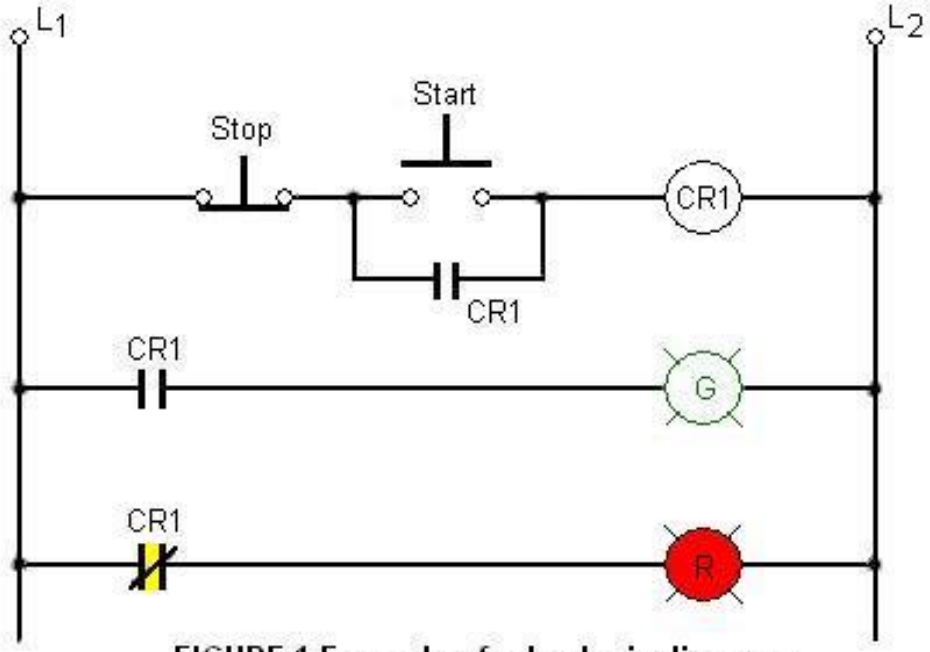
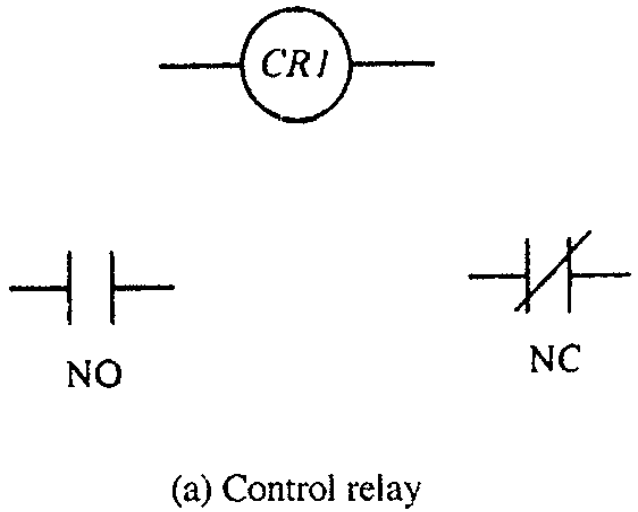


FIGURE 1 Example of relay logic diagram.

- Press **START** → Control relay (CR1) energizes.
- **Green light ON**, **Red light OFF**.
- Press **STOP** → Relay de-energizes.
- **Red light ON**, **Green light OFF**.

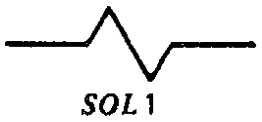
Component Representation in Ladder Diagrams



(a) Control relay



(a) Motor



(b) Solenoid



(c) Light (red)



a) Momentary action push button



b) Physical limit



c) Pressure limit

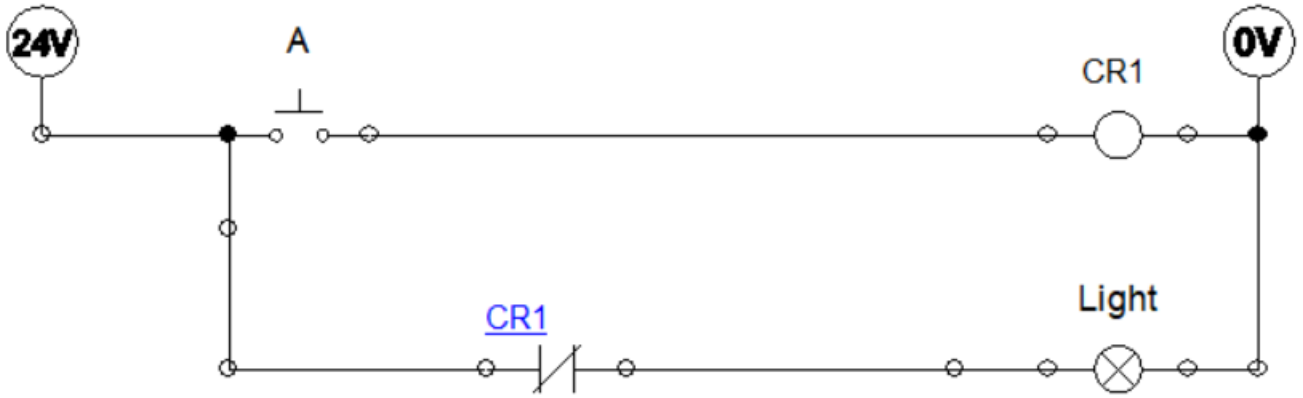


d) Temperature limit



e) Level limit

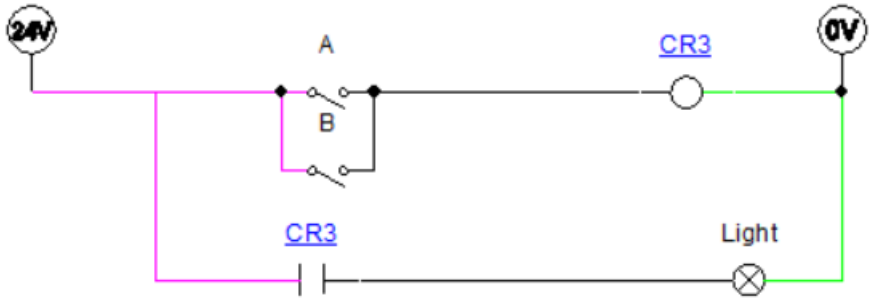
NOT Logic using Relay



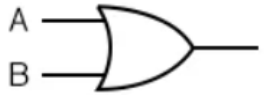
A	CR1	Output
0	0	1
1	1	0



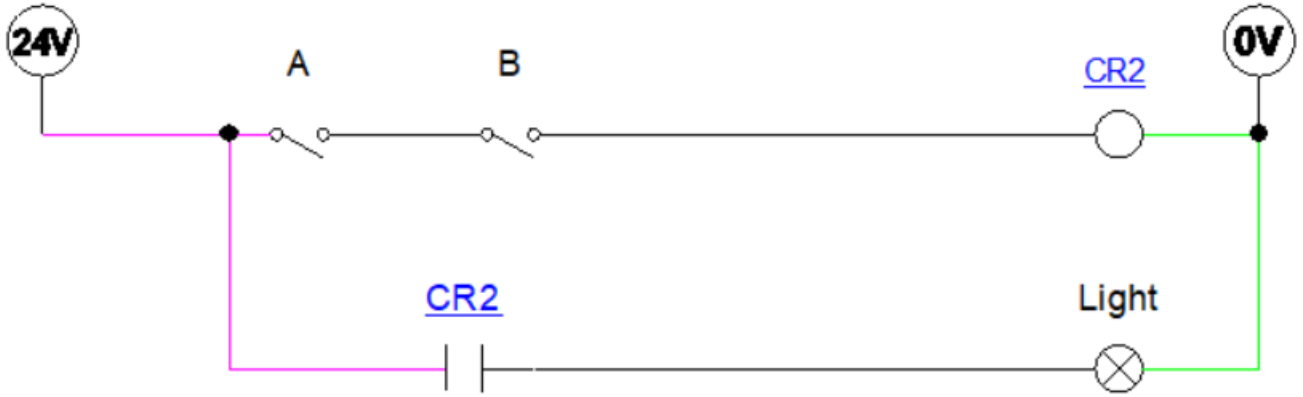
OR Logic



A	B	Output
0	0	0
0	1	1
1	0	1
1	1	1



AND Logic



A	B	Output
0	0	0
0	1	0
1	0	0
1	1	1

Relay Logic Vs. PLC Ladder Logic

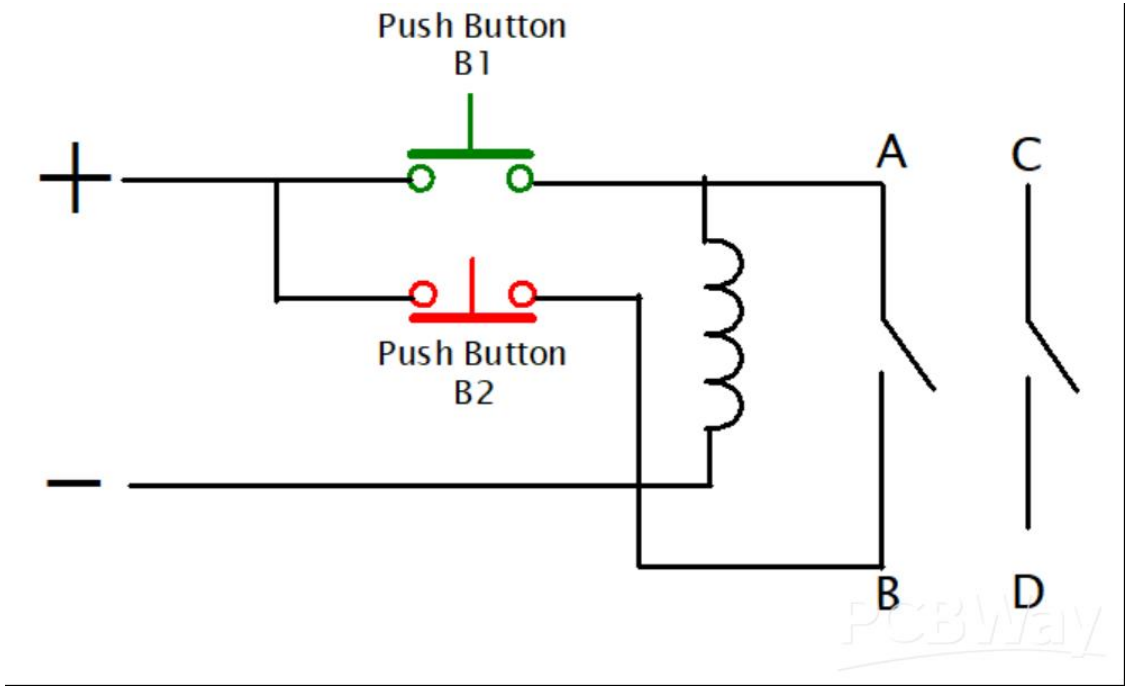
- Relay logic needs to hard wire each and every control circuit for every single control function.
- Whereas ladder logic uses the assistance of a microprocessor based device called a Programmable Logic Controller (PLC).
- Both a PLC and relay logic can perform logical computation, but a PLC does it using a microprocessor and relay logic does it using electric circuits.

Latching Relays

- Relay that maintains state after input signal is removed
- Latch Relay used as a memory element (stores ON/OFF state).

Applications:

- Motor start/stop control.
- Machine ON/OFF pushbutton stations.
- Basis for relay sequencers and PLC logic.



Control Relays – Latch Circuit

- **Start (PB1 – NO):** Energizes relay coil (RL1).
- **Self-Holding Contact:** RL1's NO contact closes → bypasses PB1 → relay stays energized even after release.
- **Stop (PB2 – NC):** Opens circuit → de-energizes coil → unlatches relay.

Indicators:
Green lamp ON when unlatched (system idle).
Red lamp ON when latched (system active)

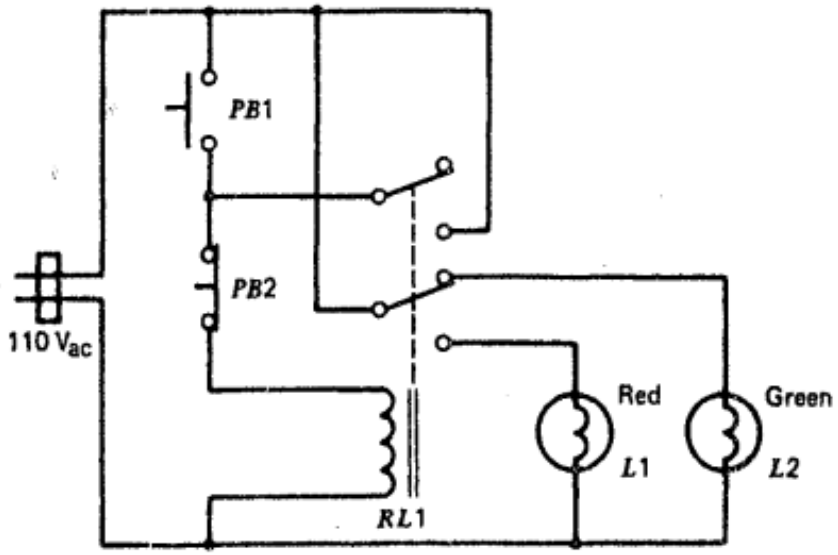


FIGURE 8.14
Use of momentary push-button switches and a relay to implement a latch.